Projection of V along W :-

If $W = [w_1, w_2, \ldots, w_r]$, where w_i's formed an orthogonal set. Then $proj(V, W) = c_1 w_1 + c_2 w_2 + \ldots + c_r w_r$ read as projection of V along W, where $c_i = \frac{\langle v, w_i \rangle}{\langle w_i, w_i \rangle}$

Ex:- Find projection of V along S.

$S = \{(1,2,1), (2,1,-4), (3,-2,1)\}$

$V = \{(7,1,9)\}$

$$\begin{aligned} Proj(V,S) &= c_1 u_1 + c_2 u_2 + c_3 u_3 \\ &= 3(1,2,1) + (-1)(2,1,-4) + 2(3,-2,1) \\ &= (3,6,3) + (-2,-1,4) + (6,-4,2) \\ &= (7,1,9) \end{aligned}$$

Gram-Schmidt Process :-

Let $S = \{u_1, u_2, \ldots, u_n\}$ is a basis of an inner product space V. Now we can construct an orthogonal basis $\{w_1, w_2, \ldots, w_n\}$ of V as follows.

Step-I $w_1 = u_1$

Step-II $w_2 = u_2 - c_1 w_1$, $c_1 = \frac{\langle u_2, w_1 \rangle}{\langle w_1, w_1 \rangle}$

$$= u_2 - \frac{\langle u_2, w_1 \rangle}{\langle w_1, w_1 \rangle} \cdot w_1$$

$$\langle w_1, w_2 \rangle = \left\langle u_1, \; u_2 - \frac{\langle u_2, w_1 \rangle}{\langle w_1, w_1 \rangle} \cdot w_1 \right\rangle = 0$$

P Kalika Maths